CSVInspect

2025-02-27

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
vehicleData = read.csv("vehicles.csv")
emissionsData = read.csv("1970-2022StateCO2Emissions.csv")

head(vehicleData)
```

```
##   barrels08 barrelsA08 charge120 charge240 city08 city08U cityA08 cityA08U
## 1  14.16714          0         0         0     19       0       0        0
## 2  27.04636          0         0         0      9       0       0        0
## 3  11.01889          0         0         0     23       0       0        0
## 4  27.04636          0         0         0     10       0       0        0
## 5  15.65842          0         0         0     17       0       0        0
## 6  13.52318          0         0         0     21       0       0        0
##   cityCD cityE cityUF co2 co2A co2TailpipeAGpm co2TailpipeGpm comb08 comb08U
## 1      0     0      0  -1   -1               0       423.1905     21       0
## 2      0     0      0  -1   -1               0       807.9091     11       0
## 3      0     0      0  -1   -1               0       329.1481     27       0
## 4      0     0      0  -1   -1               0       807.9091     11       0
## 5      0     0      0  -1   -1               0       467.7368     19       0
## 6      0     0      0  -1   -1               0       403.9545     22       0
##   combA08 combA08U combE combinedCD combinedUF cylinders displ
## 1       0        0     0          0          0         4   2.0
## 2       0        0     0          0          0        12   4.9
## 3       0        0     0          0          0         4   2.2
## 4       0        0     0          0          0         8   5.2
## 5       0        0     0          0          0         4   2.2
## 6       0        0     0          0          0         4   1.8
##                        drive engId   eng_dscr feScore fuelCost08 fuelCostA08
## 1           Rear-Wheel Drive  9011      (FFS)      -1       2150           0
## 2           Rear-Wheel Drive 22020  (GUZZLER)      -1       4100           0
## 3          Front-Wheel Drive  2100      (FFS)      -1       1650           0
```

```
## 4           Rear-Wheel Drive  2850                -1       4100           0
## 5 4-Wheel or All-Wheel Drive 66031 (FFS,TRBO)      -1       3100           0
## 6          Front-Wheel Drive 66020      (FFS)      -1       2050           0
##   fuelType        fuelType1 ghgScore ghgScoreA highway08 highway08U highwayA08
## 1  Regular Regular Gasoline       -1        -1        25          0          0
## 2  Regular Regular Gasoline       -1        -1        14          0          0
## 3  Regular Regular Gasoline       -1        -1        33          0          0
## 4  Regular Regular Gasoline       -1        -1        12          0          0
## 5  Premium Premium Gasoline       -1        -1        23          0          0
## 6  Regular Regular Gasoline       -1        -1        24          0          0
##   highwayA08U highwayCD highwayE highwayUF hlv hpv    id lv2 lv4       make
## 1           0         0        0         0   0   0     1   0   0 Alfa Romeo
## 2           0         0        0         0   0   0    10   0   0    Ferrari
## 3           0         0        0         0  19  77   100   0   0      Dodge
## 4           0         0        0         0   0   0  1000   0   0      Dodge
## 5           0         0        0         0   0   0 10000   0  14     Subaru
## 6           0         0        0         0   0   0 10001   0  15     Subaru
##                 model mpgData phevBlended pv2 pv4 range rangeCity rangeCityA
## 1  Spider Veloce 2000       Y       false   0   0     0         0          0
## 2          Testarossa       N       false   0   0     0         0          0
## 3             Charger       Y       false   0   0     0         0          0
## 4 B150/B250 Wagon 2WD       N       false   0   0     0         0          0
## 5    Legacy AWD Turbo       N       false   0  90     0         0          0
## 6              Loyale       N       false   0  88     0         0          0
##   rangeHwy rangeHwyA           trany   UCity UCityA UHighway UHighwayA
## 1        0         0    Manual 5-spd 23.3333      0  35.0000         0
## 2        0         0    Manual 5-spd 11.0000      0  19.0000         0
## 3        0         0    Manual 5-spd 29.0000      0  47.0000         0
## 4        0         0 Automatic 3-spd 12.2222      0  16.6667         0
## 5        0         0    Manual 5-spd 21.0000      0  32.0000         0
## 6        0         0 Automatic 3-spd 27.0000      0  33.0000         0
##            VClass year youSaveSpend       baseModel guzzler trans_dscr tCharger
## 1     Two Seaters 1985        -2750          Spider                         NA
## 2     Two Seaters 1985       -12500      Testarossa       T                 NA
## 3 Subcompact Cars 1985         -250         Charger                SIL       NA
## 4            Vans 1985       -12500 B150/B250 Wagon                         NA
## 5    Compact Cars 1993        -7500  Legacy/Outback                        TRUE
## 6    Compact Cars 1993        -2250          Loyale                         NA
##   sCharger atvType fuelType2 rangeA evMotor mfrCode c240Dscr charge240b
## 1                                                                    0
## 2                                                                    0
## 3                                                                    0
## 4                                                                    0
## 5                                                                    0
## 6                                                                    0
##   c240bDscr                    createdOn                   modifiedOn startStop
## 1           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
## 2           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
## 3           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
## 4           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
## 5           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
## 6           Tue Jan 01 00:00:00 EST 2013 Tue Jan 01 00:00:00 EST 2013
##   phevCity phevHwy phevComb
## 1        0       0        0
```

```
## 2        0       0        0
## 3        0       0        0
## 4        0       0        0
## 5        0       0        0
## 6        0       0        0
```

```r
head(emissionsData)
```

```
##        State X1970 X1971 X1972 X1973 X1974 X1975 X1976 X1977 X1978 X1979 X1980
## 1    Alabama 102.7  98.5 105.0 109.6 108.8 107.8 108.1 111.7 106.7 111.6 107.2
## 2     Alaska  11.4  12.6  13.4  12.5  12.8  14.5  16.0  18.0  19.5  17.5  17.4
## 3    Arizona  24.9  27.0  30.3  34.5  36.8  38.2  43.8  50.5  49.3  56.2  52.9
## 4   Arkansas  36.2  35.1  37.2  40.9  39.2  36.4  38.9  41.7  42.5  40.3  37.5
## 5 California 294.7 306.1 313.0 329.5 304.7 311.7 327.1 354.7 345.4 362.1 344.4
## 6   Colorado  43.1  43.6  47.5  51.1  50.5  51.8  55.2  58.3  58.5  58.9  58.4
##   X1981 X1982 X1983 X1984 X1985 X1986 X1987 X1988 X1989 X1990 X1991 X1992 X1993
## 1 103.8  91.0  90.0  95.4 101.8 101.9 104.0 105.2 110.0 109.8 114.1 121.0 125.3
## 2  17.1  24.0  26.0  28.7  29.0  31.4  29.8  30.3  33.3  34.0  34.5  35.9  35.7
## 3  59.9  58.5  54.2  58.5  61.1  56.3  56.4  59.6  65.5  63.1  64.0  66.8  69.1
## 4  43.0  42.8  47.1  45.0  49.2  50.0  47.1  51.0  51.3  50.8  49.8  51.4  50.4
## 5 334.5 299.1 293.8 315.8 320.7 306.7 335.9 344.8 359.5 360.3 349.0 352.8 342.1
## 6  57.4  58.6  56.8  60.5  61.2  60.4  61.0  63.3  64.5  65.8  67.5  68.5  72.0
##   X1994 X1995 X1996 X1997 X1998 X1999 X2000 X2001 X2002 X2003 X2004 X2005 X2006
## 1 123.2 131.0 137.2 134.1 133.5 135.8 142.3 133.3 138.3 139.8 142.0 143.5 145.8
## 2  35.5  40.0  41.1  41.2  42.2  42.8  43.6  42.5  42.7  42.9  46.2  47.5  45.4
## 3  71.8  66.7  68.6  71.7  76.7  80.6  86.6  88.9  88.3  90.5  97.3  97.3 100.5
## 4  54.3  57.7  60.2  59.1  60.5  62.7  63.3  62.5  61.2  62.2  62.5  60.2  62.1
## 5 359.0 348.5 349.5 352.3 362.2 365.6 382.0 385.3 383.7 374.0 392.0 388.8 397.5
## 6  72.6  72.7  75.6  75.8  78.1  80.2  85.3  93.0  91.3  90.7  93.2  95.4  96.3
##   X2007 X2008 X2009 X2010 X2011 X2012 X2013 X2014 X2015 X2016 X2017 X2018 X2019
## 1 147.3 139.4 119.7 132.4 129.4 122.4 120.1 122.3 118.8 113.4 108.1 111.9 105.8
## 2  43.7  39.1  37.2  37.1  37.0  36.1  34.0  33.8  34.9  33.3  33.6  34.4  34.2
## 3 102.3 102.6  93.7  99.4  97.5  95.3  99.0  97.0  94.6  90.4  90.0  93.6  92.1
## 4  63.4  64.2  61.4  66.0  67.3  66.1  68.2  68.7  58.8  61.8  63.8  70.5  64.8
## 5 402.3 383.8 369.8 356.1 341.8 347.9 348.3 344.0 349.7 351.2 354.5 356.7 356.5
## 6  98.9  97.0  92.8  95.5  92.1  90.8  91.7  92.3  90.8  87.9  88.2  89.4  91.1
##   X2020 X2021 X2022 Percent Absolute Percent.1 Absolute.1
## 1  98.0 108.3 109.3    6.4%      6.6      0.9%        1.0
## 2  35.9  39.7  41.3  263.6%     29.9      4.0%        1.6
## 3  79.7  82.6  81.4  226.5%     56.5     -1.5%       -1.2
## 4  54.4  61.7  63.2   74.4%     26.9      2.4%        1.5
## 5 302.0 324.1 326.2   10.7%     31.5      0.7%        2.1
## 6  79.3  84.7  88.7  106.0%     45.6      4.8%        4.0
```

```r
# the vehicle data set has a lot of extra information, lets only keep the relevant information
vehicleData2022 = vehicleData %>%
  select(city08U, cityA08U, highway08U, highwayA08U, comb08U, combA08U, co2TailpipeGpm, co2TailpipeAGpm
  filter(year == 2022 )


colnames(vehicleData2022)[1] = "SFVCityMPG"
colnames(vehicleData2022)[2] = "DFVCityMPG"
colnames(vehicleData2022)[3] = "SFVHighwayMPG"
```

```r
colnames(vehicleData2022)[4] = "DFVHighwayMPG"
colnames(vehicleData2022)[5] = "SFCombinedMPG"
colnames(vehicleData2022)[6] = "DFVCombinedMPG"
colnames(vehicleData2022)[7] = "SFVCO2Gpm"
colnames(vehicleData2022)[8] = "DFVCO2Gpm"


# lets get the electric vehicles from this data set, which would be a 0 for tailpipe emissions in both
electricVehicles2022 = vehicleData2022[vehicleData2022$fuelType == "Electricity", ]

# get relevant data for only electric vehicles with empty columns that will be filled via the state dat
electricVehicles2022 = electricVehicles2022 %>%
  select(make, model, year, fuelType, cityE, highwayE, combE, SFVCO2Gpm, SFVCityMPG, SFVHighwayMPG, SFC

# the cityE, highwayE, and combE are in kw-hours per 100 miles, but I want to know the effect of 1 kWh
electricVehicles2022 = electricVehicles2022 %>%
  mutate(cityMpkWh = 100 / cityE) %>%
  mutate(highwayMpKwh = 100 / highwayE) %>%
  mutate(combMpKwh = 100 / combE)

# now that that is done, lets make our final table with only the information that we need
electricVehicles2022 = electricVehicles2022 %>%
  select(make, model, year, fuelType, cityMpkWh, highwayMpKwh, combMpKwh, SFVCO2Gpm)
colnames(electricVehicles2022)[8] = "gCo2pM"


# now we can go back and refine our data for non electric vehicles
nonElectricVehicleData2022 = vehicleData2022[vehicleData2022$fuelType != "Electricity", ]

# dropping the dual fuel grams of co2 per mile since it is rarely used only 17 times, we will use the v
# also going to remove the hydrogen vehicles as we have not accounted for pollution from hydrogen acqui
nonElectricVehicleData2022 = nonElectricVehicleData2022 %>%
  select(make, model, year, fuelType, SFVCityMPG, SFVHighwayMPG, SFCombinedMPG, SFVCO2Gpm) %>%
  filter(fuelType != "Hydrogen")

colnames(nonElectricVehicleData2022)[5] = "cityMPG"
colnames(nonElectricVehicleData2022)[6] = "highwayMPG"
colnames(nonElectricVehicleData2022)[7] = "combinedMPG"
colnames(nonElectricVehicleData2022)[8] = "gCo2pM"
```

```r
# clean and mutate the data to get the information we want

emissionsData$X2022 = as.numeric(as.character(emissionsData$X2022))
```

```
## Warning: NAs introduced by coercion
```

```r
stateData2022 = emissionsData %>%
  select(State, X2022)

# now we need to convert the million tons of CO2 into grams of cO2
stateData2022 = stateData2022 %>%
  mutate(gCo2 = X2022 * 1.1e7) %>%
  mutate(gCo2pkWh = 0)
```

```r
head(stateData2022)
```

```
##        State X2022       gCo2 gCo2pkWh
## 1    Alabama 109.3 1202300000        0
## 2     Alaska  41.3  454300000        0
## 3    Arizona  81.4  895400000        0
## 4   Arkansas  63.2  695200000        0
## 5 California 326.2 3588200000        0
## 6   Colorado  88.7  975700000        0
```

```r
colnames(stateData2022)[2] = "millionTonsOfCO2"

head(stateData2022)
```

```
##        State millionTonsOfCO2       gCo2 gCo2pkWh
## 1    Alabama            109.3 1202300000        0
## 2     Alaska             41.3  454300000        0
## 3    Arizona             81.4  895400000        0
## 4   Arkansas             63.2  695200000        0
## 5 California            326.2 3588200000        0
## 6   Colorado             88.7  975700000        0
```

```r
# data set form the eia for electricity generation from 2022

electricityGeneration2022 = read.csv("stateGeneration2022.csv")


electricityGeneration2022 = electricityGeneration2022 %>%
  mutate(netKWhGeneration = NetGeneration.MWh. * 1000)

head(electricityGeneration2022)
```

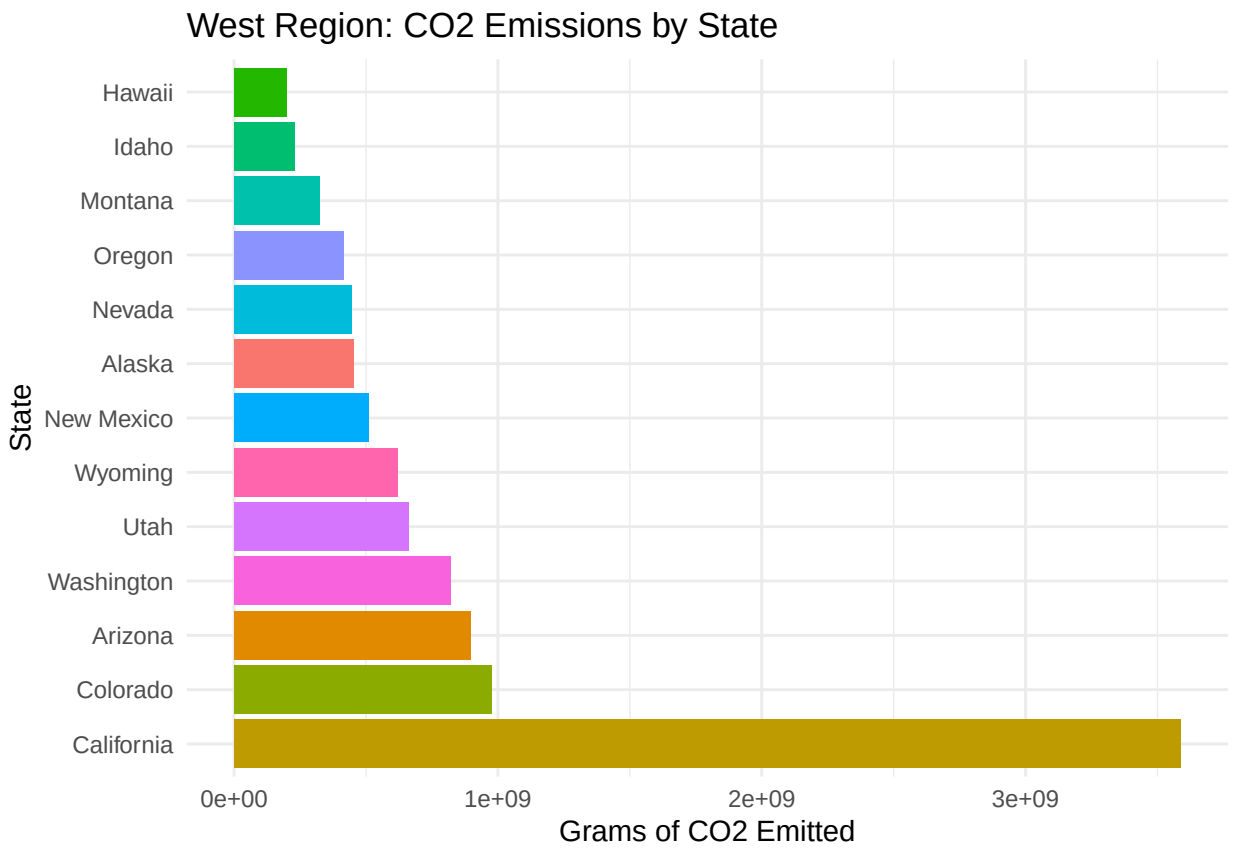
```
##        State AverageRetailPrice.cents.kWh. NetSummerCapacity.MW.
## 1    Alabama                         11.59                28910
## 2     Alaska                         20.73                 2820
## 3    Arizona                         11.31                28202
## 4   Arkansas                          9.91                14954
## 5 California                         22.33                85981
## 6   Colorado                         11.75                18092
##   NetGeneration.MWh. TotalRetailSales.MWh. netKWhGeneration
## 1          144788893              87027545     144788893000
## 2            6694128               6002080       6694128000
## 3          104698773              84196517     104698773000
## 4           65905030              48997663      65905030000
## 5          203383857             251869136     203383857000
## 6           58044009              56763041      58044009000
```

```r
# lets create some visualizations and split the states into regions
# the us census bureau divides the US into west, midwest, south and north. So we will follow this divis
westData2022 = stateData2022 %>%
```

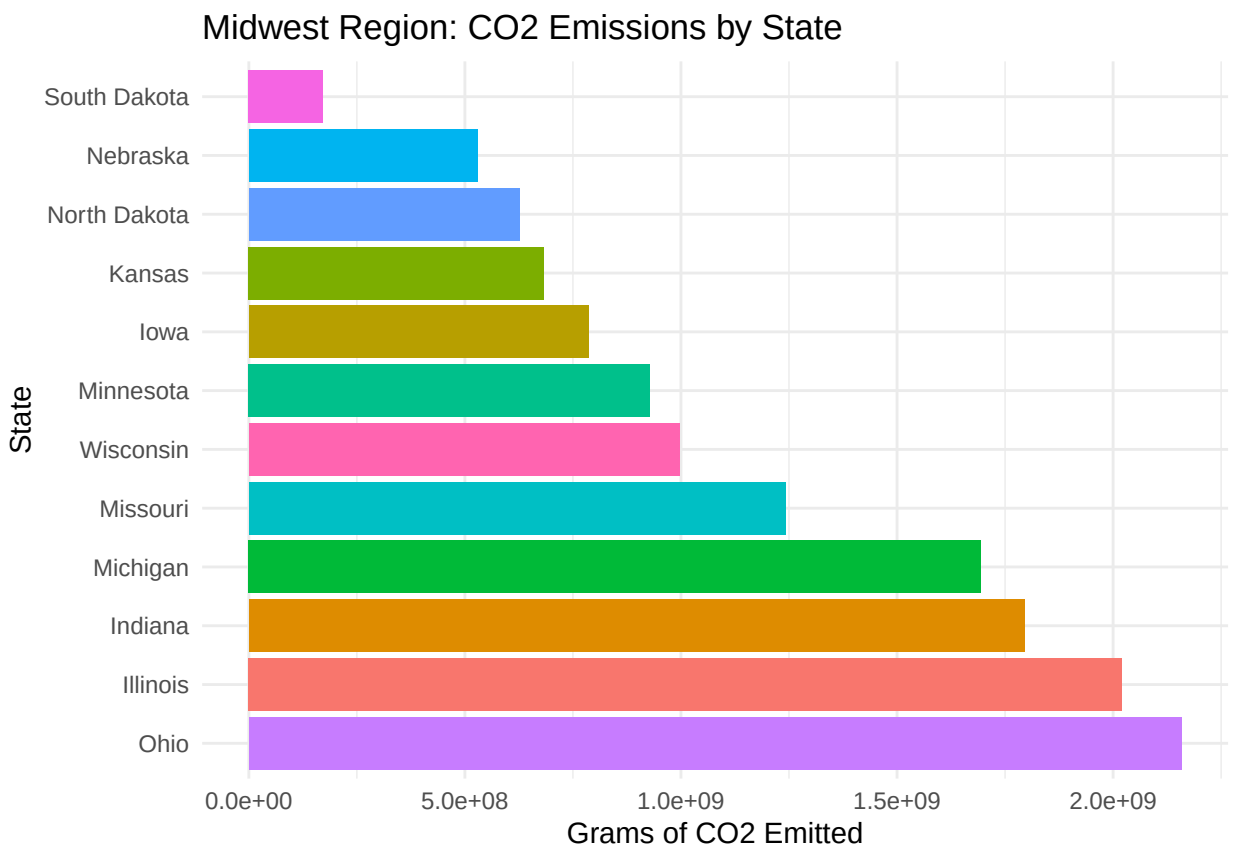
```
  filter(State %in% c("Arizona", "Colorado", "Idaho", "Montana", "Nevada",
                      "New Mexico", "Utah", "Wyoming", "Alaska", "California",
                      "Hawaii", "Oregon", "Washington"))

midWestData2022 = stateData2022 %>%
  filter(State %in% c("Illinois", "Indiana", "Michigan", "Ohio", "Wisconsin", "Iowa", "Kansas", "Minnes
                      "Nebraska", "North Dakota", "South Dakota"))

southData2022 = stateData2022 %>%
  filter(State %in% c("Delaware", "Florida", "Georgia", "Maryland", "North Carolina", "South Carolina",
                      "West Virginia", "Alabama", "Kentucky", "Mississippi", "Tennessee", "Arkansas", "
                      "Oklahoma", "Texas", "District of Columbia"))

northData2022 = stateData2022 %>%
  filter(State %in% c("Connecticut", "Maine", "Massachusetts", "New Hampshire", "Rhode Island", "Vermon
                      "New Jersey", "New York", "Pennsylvania"))

# to ensure all data was correctly divided
sum(length(westData2022$State)) + sum(length(midWestData2022$State)) + sum(length(southData2022$State))
```

```
## [1] 51
```

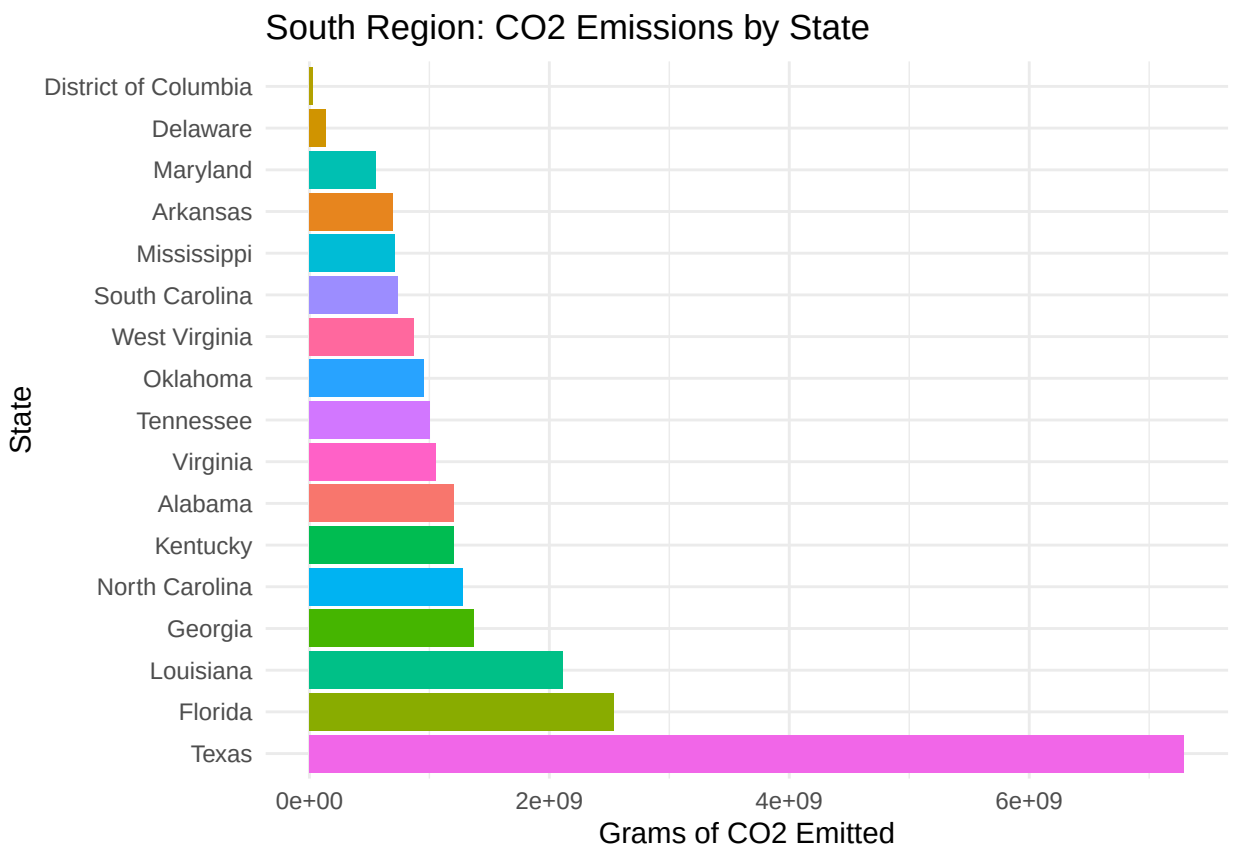
```
# so now lets create some plots of grams of co2 emitted by each region
emissionsRegionPlot <- function(data, region_name) {
  ggplot(data, aes(x = reorder(State, -gCo2), y = gCo2, fill = State)) +
    geom_bar(stat = "identity") +
    coord_flip() +  # Flip coordinates for better readability
    theme_minimal() +
    labs(title = paste(region_name, "Region: CO2 Emissions by State"),
         x = "State",
         y = "Grams of CO2 Emitted") +
    theme(legend.position = "none")  # Hide legend for better visualization
}

# Plot each region
emissionsRegionPlot(westData2022, "West")
```



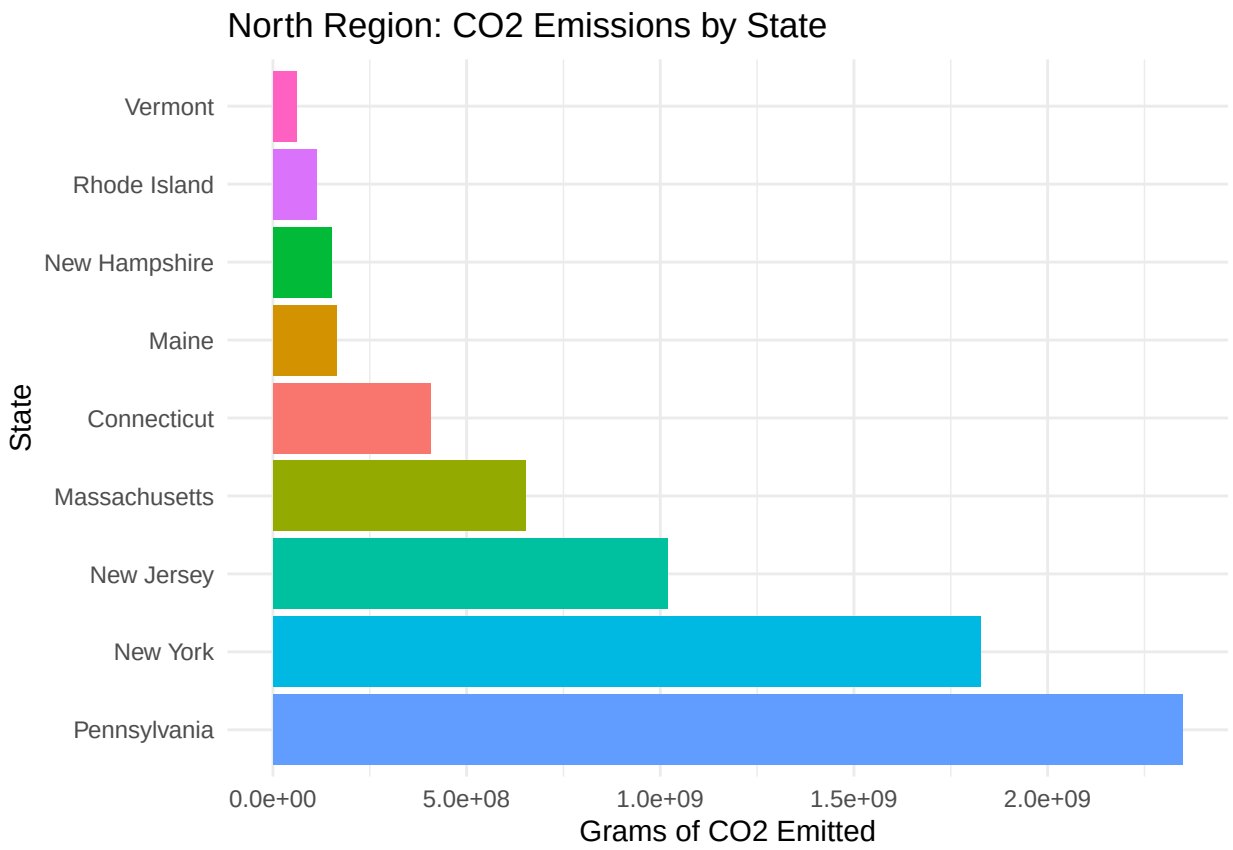

```
emissionsRegionPlot(midWestData2022, "Midwest")
```




```
emissionsRegionPlot(southData2022, "South")
```



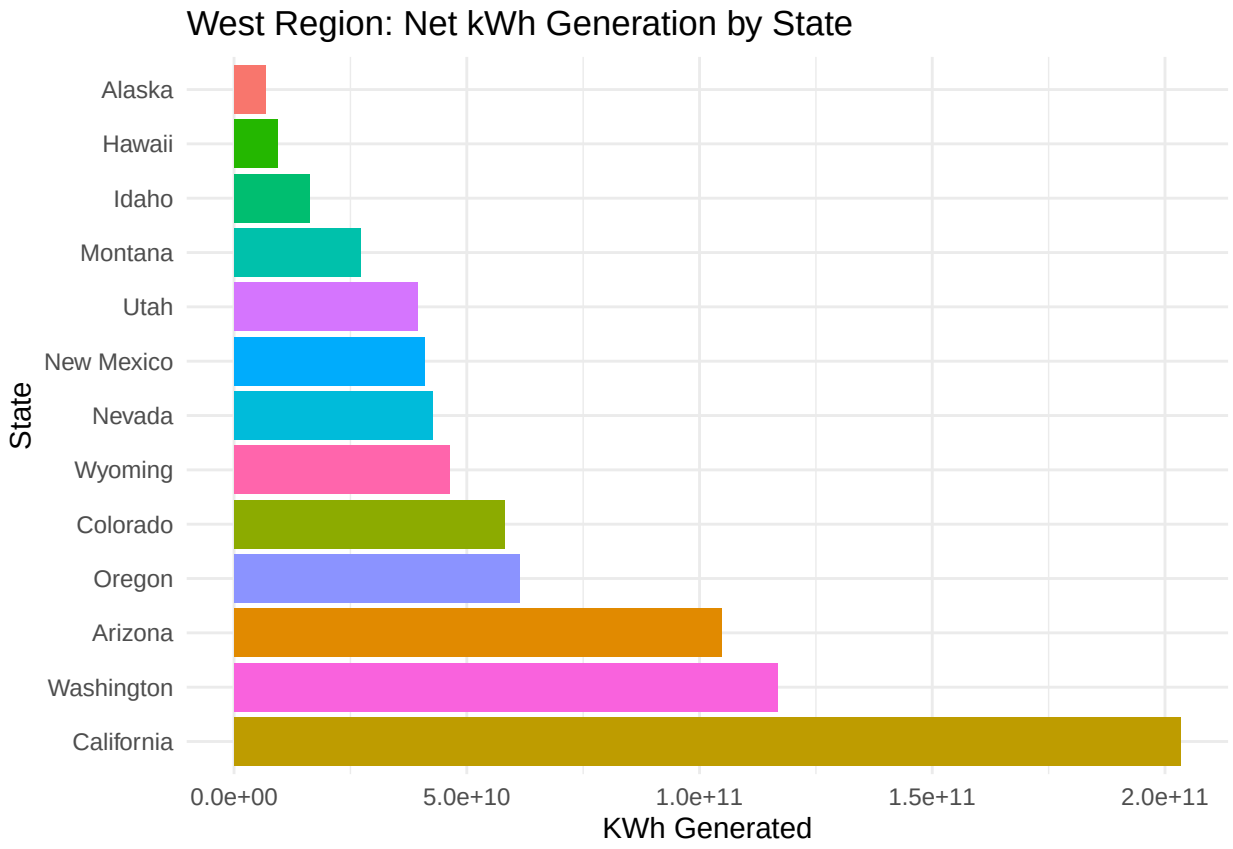

```
emissionsRegionPlot(northData2022, "North")
```



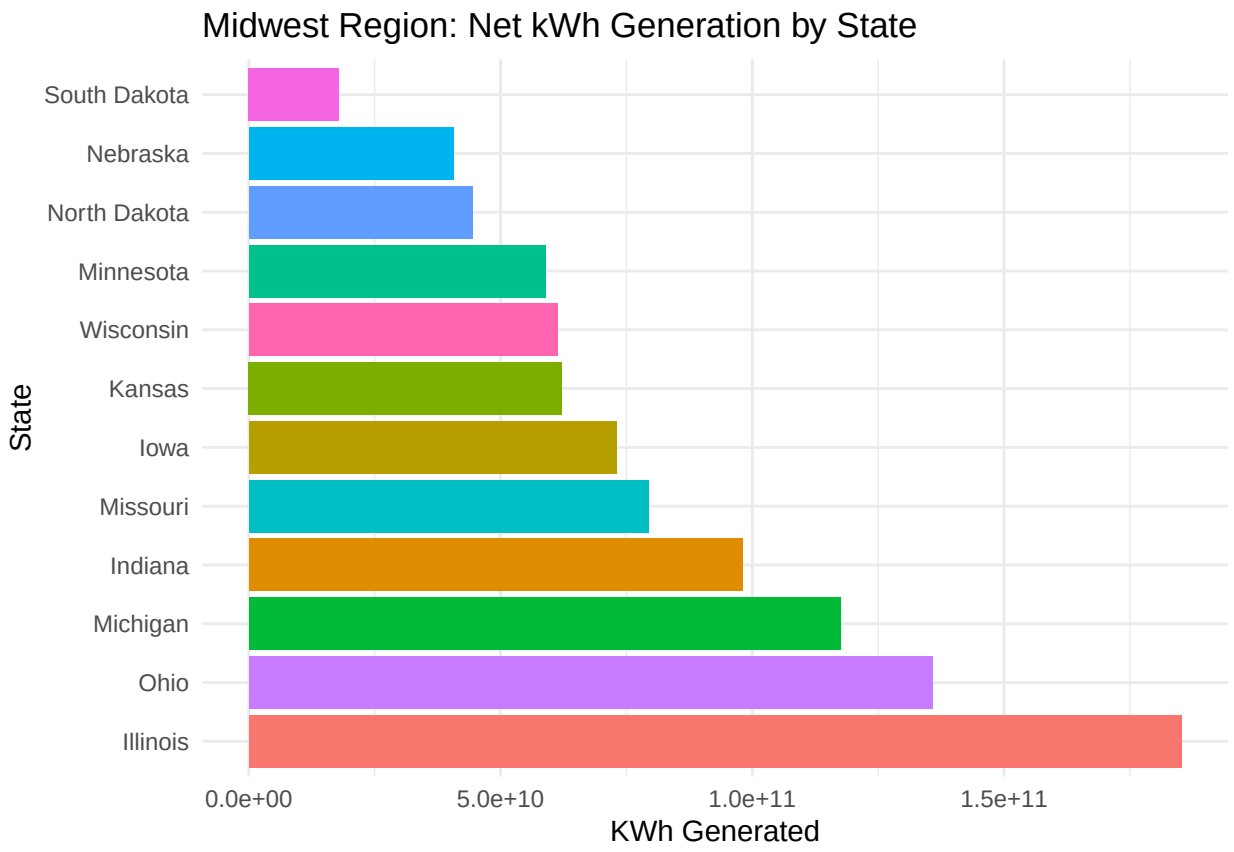

```
# now I am going to do the same thing but with electricity generation
# splitting these into the same regions above
westGenerationData2022 = electricityGeneration2022 %>%
  filter(State %in% c("Arizona", "Colorado", "Idaho", "Montana", "Nevada",
                      "New Mexico", "Utah", "Wyoming", "Alaska", "California",
                      "Hawaii", "Oregon", "Washington"))

midWestGenerationData2022 = electricityGeneration2022 %>%
  filter(State %in% c("Illinois", "Indiana", "Michigan", "Ohio", "Wisconsin", "Iowa", "Kansas", "Minnes
                      "Nebraska", "North Dakota", "South Dakota"))

southGenerationData2022 = electricityGeneration2022 %>%
  filter(State %in% c("Delaware", "Florida", "Georgia", "Maryland", "North Carolina", "South Carolina",
                      "West Virginia", "Alabama", "Kentucky", "Mississippi", "Tennessee", "Arkansas", "
                      "Oklahoma", "Texas", "District of Columbia"))

northGenerationData2022 = electricityGeneration2022 %>%
  filter(State %in% c("Connecticut", "Maine", "Massachusetts", "New Hampshire", "Rhode Island", "Vermon
                      "New Jersey", "New York", "Pennsylvania"))

# to ensure all data was correctly divided
sum(length(westGenerationData2022$State)) + sum(length(midWestGenerationData2022$State)) + sum(length(s
```

```
## [1] 51
```

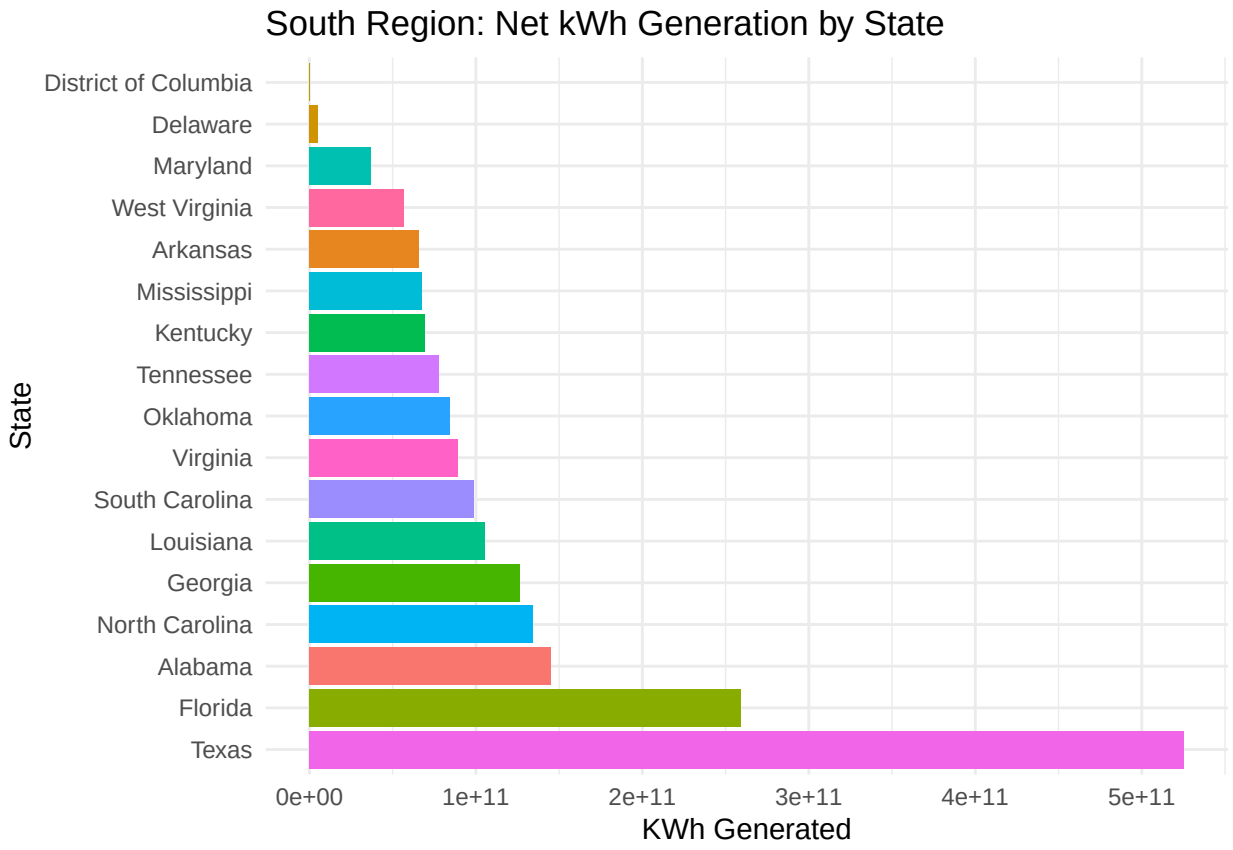
```r
# now I can create the same set of plots as above, but for electricity generation
generationRegionPlot = function(data, region_name) {
  ggplot(data, aes(x = reorder(State, -netKWhGeneration), y = netKWhGeneration, fill = State)) +
    geom_bar(stat = "identity") +
    coord_flip() +  # Flip coordinates for better readability
    theme_minimal() +
    labs(title = paste(region_name, "Region: Net kWh Generation by State"),
         x = "State",
         y = "KWh Generated") +
    theme(legend.position = "none")  # Hide legend for better visualization
}

# Plot each region
generationRegionPlot(westGenerationData2022, "West")
```



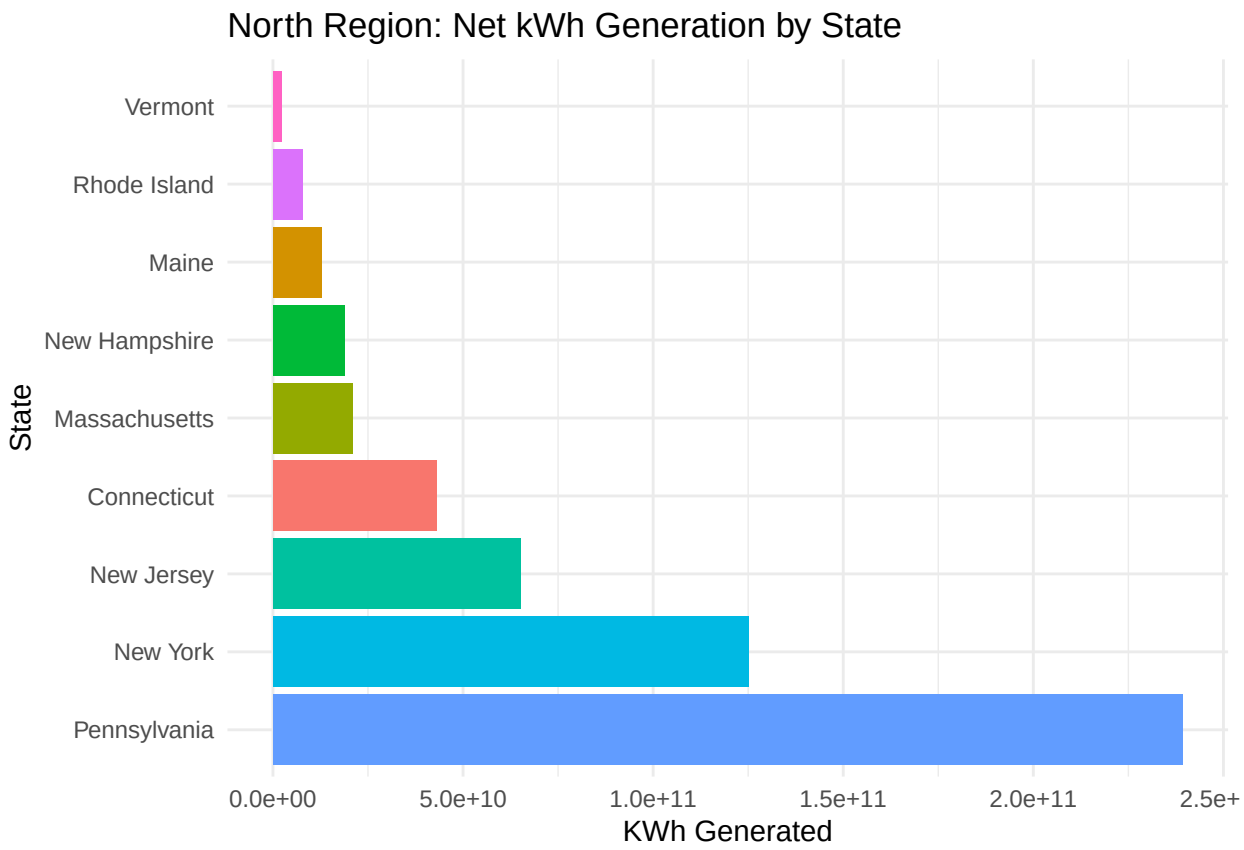

```r
generationRegionPlot(midWestGenerationData2022, "Midwest")
```



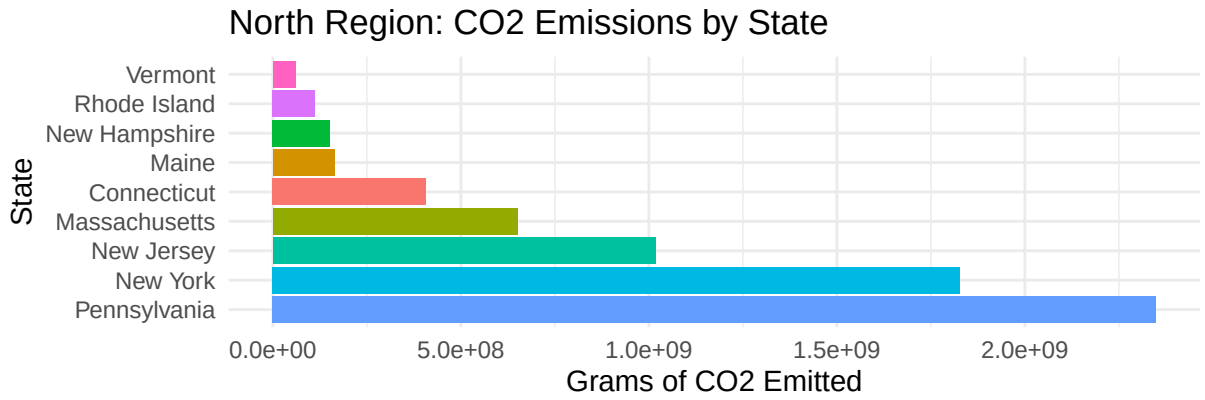

```
generationRegionPlot(southGenerationData2022, "South")
```

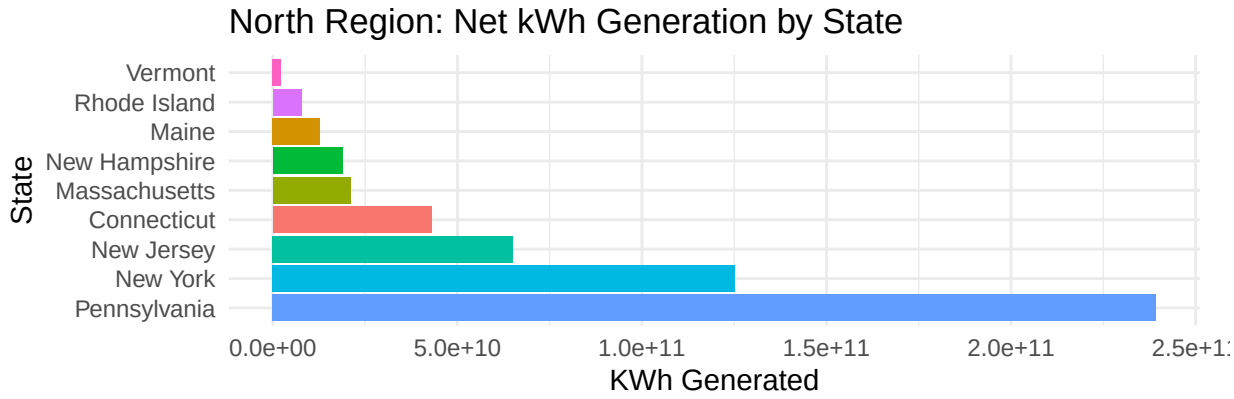



```
generationRegionPlot(northGenerationData2022, "North")
```

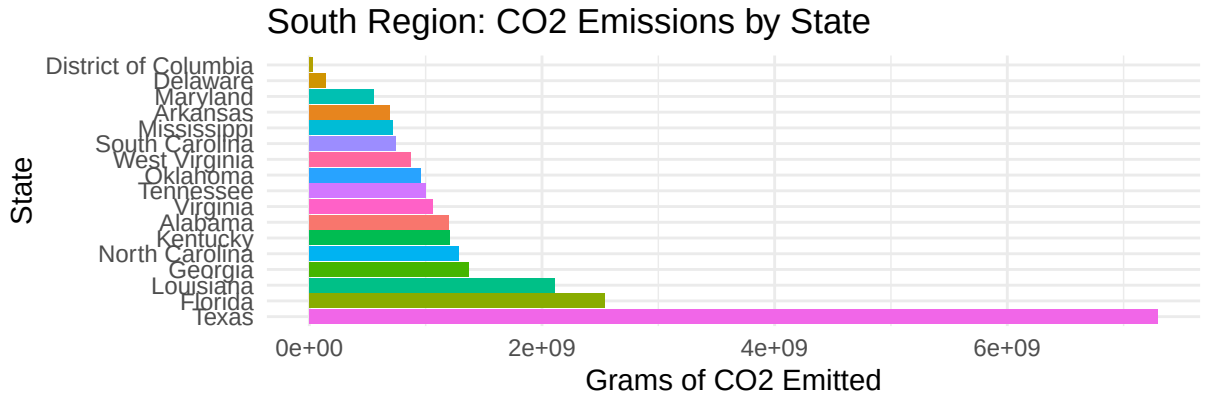


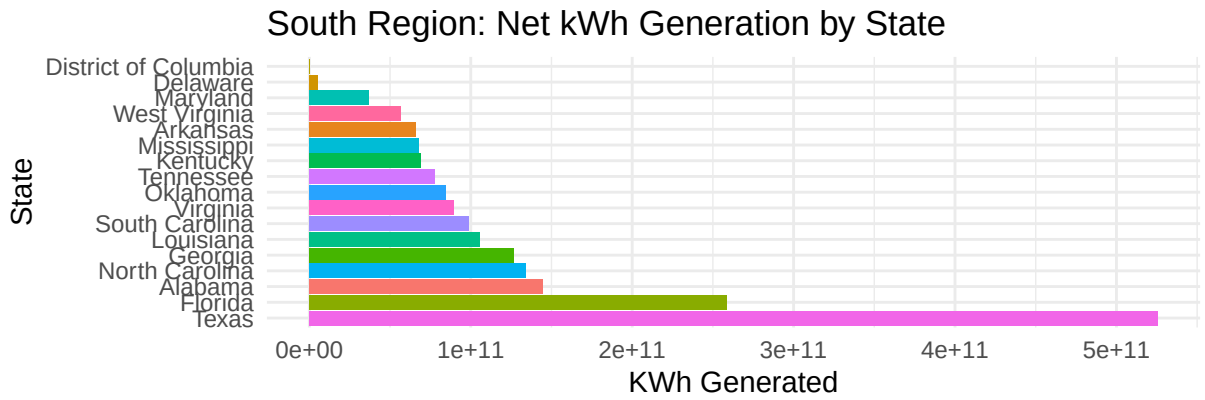

```
# now plotting side by side by region
library(patchwork)

emissionsRegionPlot(northData2022, "North") / generationRegionPlot(northGenerationData2022, "North")
```



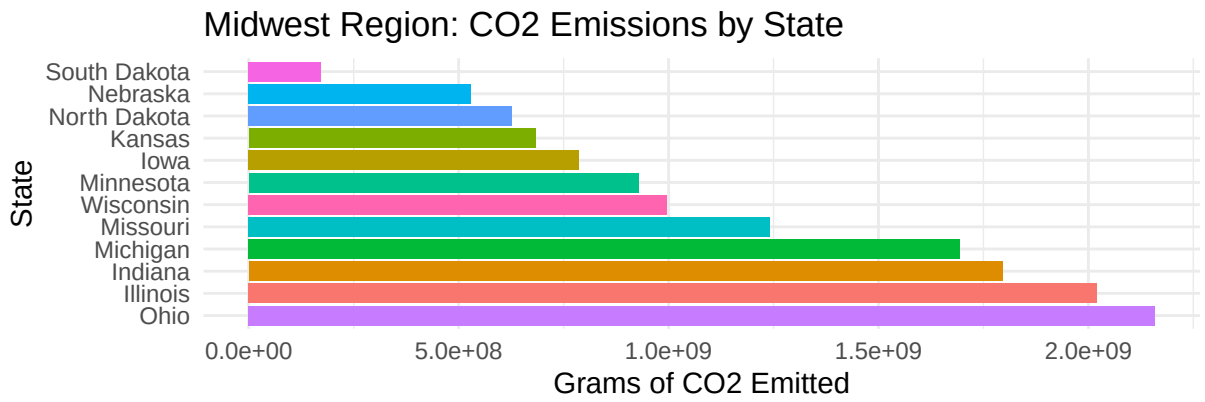


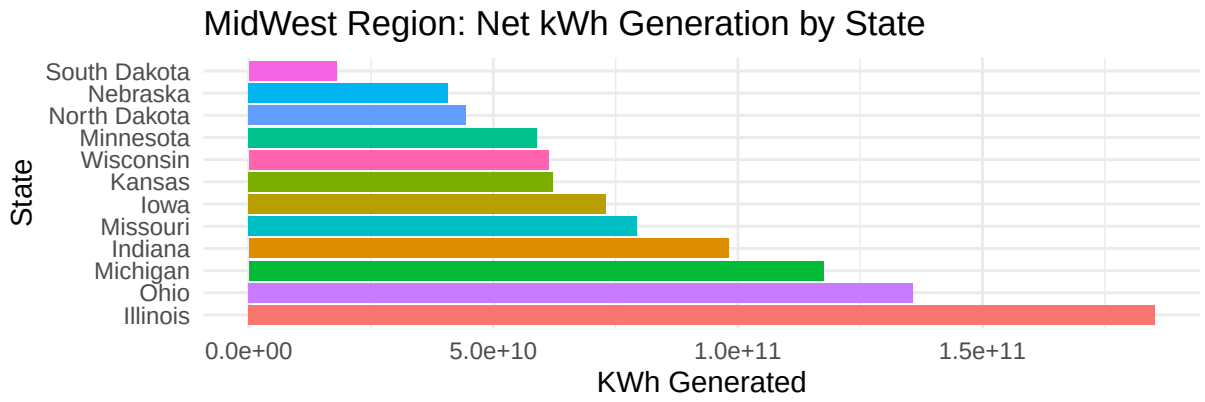

```
emissionsRegionPlot(southData2022, "South") / generationRegionPlot(southGenerationData2022, "South")
```



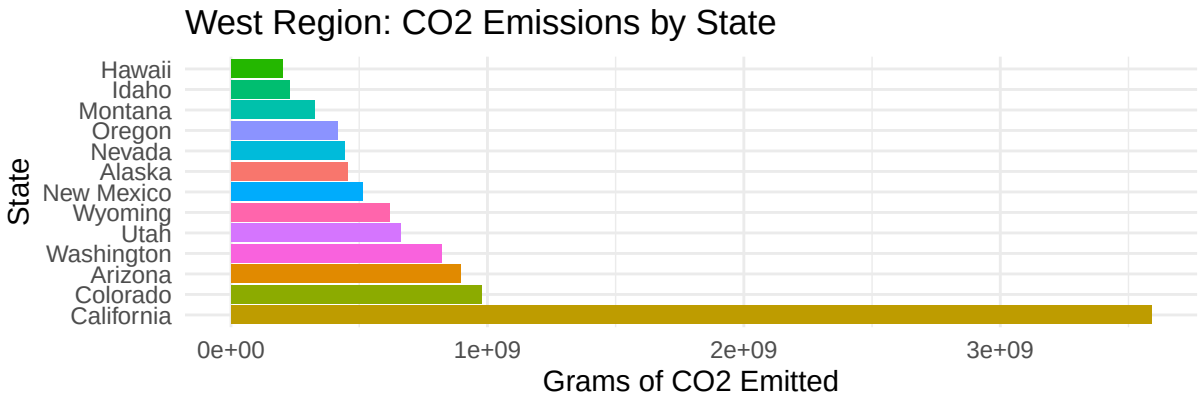


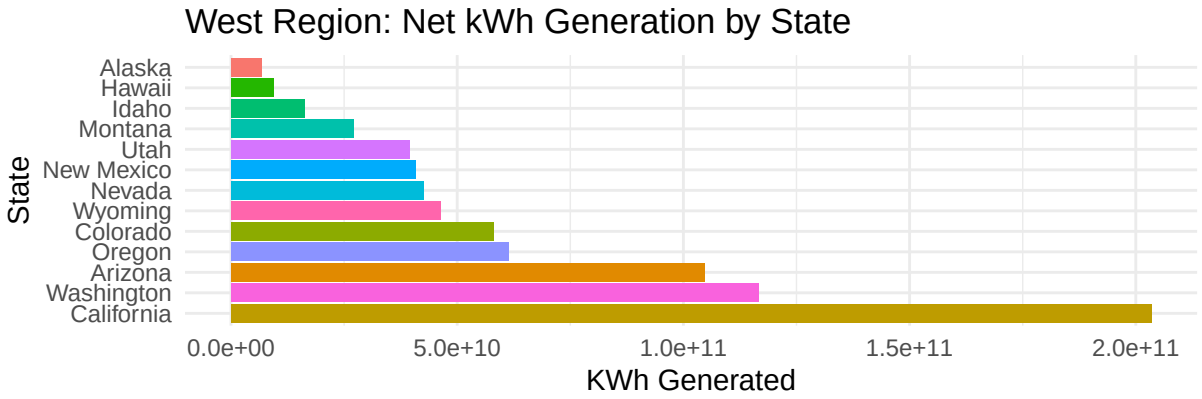
```
emissionsRegionPlot(midWestData2022, "Midwest") / generationRegionPlot(midWestGenerationData2022, "MidW
```






```
emissionsRegionPlot(westData2022, "West") / generationRegionPlot(westGenerationData2022, "West")
```






```
# now that I have these graphs split up and visualized, I really need to join the generation and emissi
finalElectricityData2022 = merge(stateData2022, electricityGeneration2022, by="State")
head(finalElectricityData2022)
```

```
##        State millionTonsOfCO2       gCo2 gCo2pkWh AverageRetailPrice.cents.kWh.
## 1    Alabama            109.3 1202300000        0                         11.59
## 2     Alaska             41.3  454300000        0                         20.73
## 3    Arizona             81.4  895400000        0                         11.31
## 4   Arkansas             63.2  695200000        0                          9.91
## 5 California            326.2 3588200000        0                         22.33
## 6   Colorado             88.7  975700000        0                         11.75
##   NetSummerCapacity.MW. NetGeneration.MWh. TotalRetailSales.MWh.
## 1                 28910          144788893              87027545
## 2                  2820            6694128               6002080
## 3                 28202          104698773              84196517
## 4                 14954           65905030              48997663
## 5                 85981          203383857             251869136
## 6                 18092           58044009              56763041
##   netKWhGeneration
## 1     144788893000
## 2       6694128000
## 3     104698773000
## 4      65905030000
## 5     203383857000
## 6      58044009000
```

```r
# now we can get the value for the grams of c02 per kilowatt hour for each state
finalElectricityData2022 = finalElectricityData2022 %>%
  mutate(gCo2pkWh = gCo2 / netKWhGeneration)

head(finalElectricityData2022)
```

```
##        State millionTonsOfCO2       gCo2    gCo2pkWh
## 1    Alabama            109.3 1202300000 0.008303814
## 2     Alaska             41.3  454300000 0.067865449
## 3    Arizona             81.4  895400000 0.008552154
## 4   Arkansas             63.2  695200000 0.010548512
## 5 California            326.2 3588200000 0.017642501
## 6   Colorado             88.7  975700000 0.016809659
##   AverageRetailPrice.cents.kWh. NetSummerCapacity.MW. NetGeneration.MWh.
## 1                         11.59                28910          144788893
## 2                         20.73                 2820            6694128
## 3                         11.31                28202          104698773
## 4                          9.91                14954           65905030
## 5                         22.33                85981          203383857
## 6                         11.75                18092           58044009
##   TotalRetailSales.MWh. netKWhGeneration
## 1              87027545     144788893000
## 2               6002080       6694128000
## 3              84196517     104698773000
## 4              48997663      65905030000
## 5             251869136     203383857000
## 6              56763041      58044009000
```

```r
# now I am going to rearrange the columns
finalElectricityData2022 = finalElectricityData2022 %>%
  select(State, millionTonsOfCO2, gCo2, gCo2pkWh, netKWhGeneration, AverageRetailPrice.cents.kWh., Total
         NetGeneration.MWh., NetSummerCapacity.MW.)

head(finalElectricityData2022)
```

```
##        State millionTonsOfCO2       gCo2    gCo2pkWh netKWhGeneration
## 1    Alabama            109.3 1202300000 0.008303814     144788893000
## 2     Alaska             41.3  454300000 0.067865449       6694128000
## 3    Arizona             81.4  895400000 0.008552154     104698773000
## 4   Arkansas             63.2  695200000 0.010548512      65905030000
## 5 California            326.2 3588200000 0.017642501     203383857000
## 6   Colorado             88.7  975700000 0.016809659      58044009000
##   AverageRetailPrice.cents.kWh. TotalRetailSales.MWh. NetGeneration.MWh.
## 1                         11.59              87027545          144788893
## 2                         20.73               6002080            6694128
## 3                         11.31              84196517          104698773
## 4                          9.91              48997663           65905030
## 5                         22.33             251869136          203383857
## 6                         11.75              56763041           58044009
##   NetSummerCapacity.MW.
## 1                 28910
## 2                  2820
## 3                 28202
```

```
## 4                 14954
## 5                 85981
## 6                 18092
```

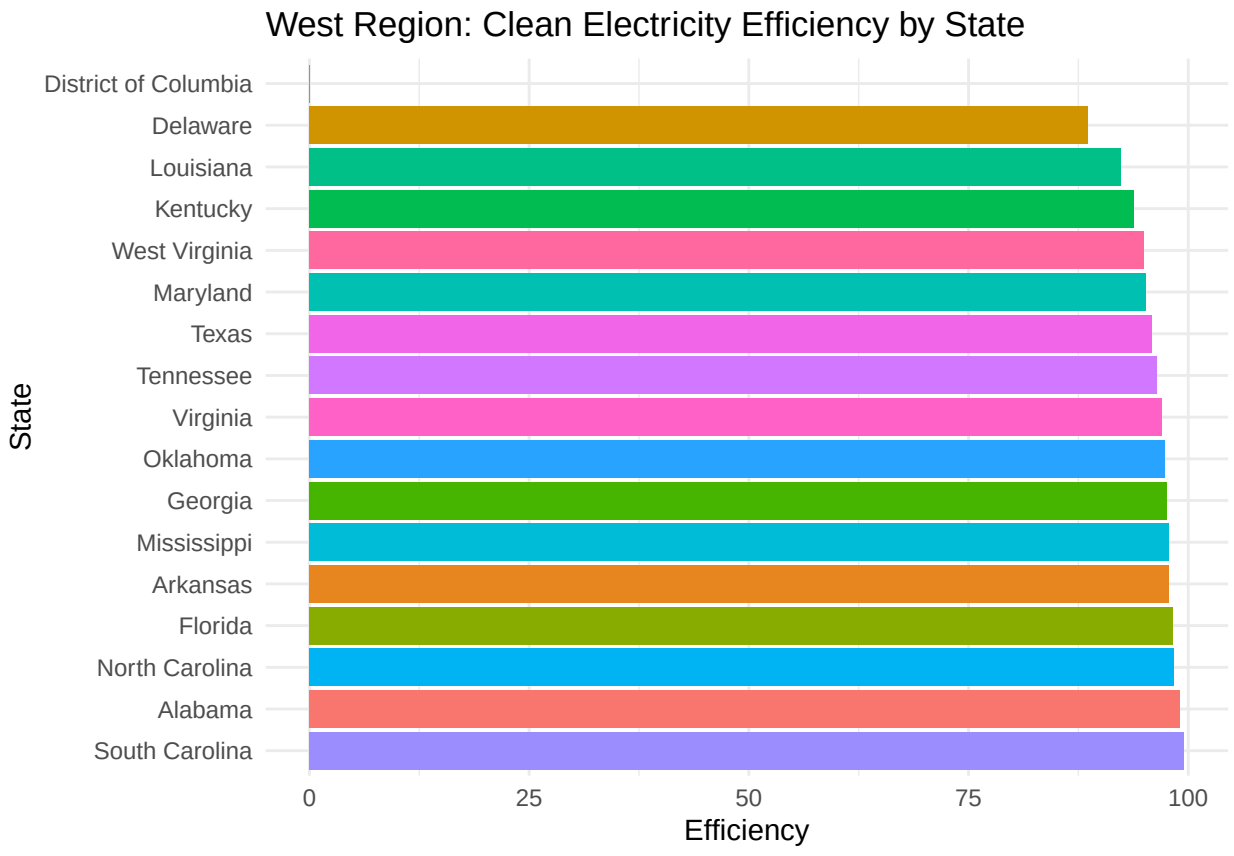
```
# so now I would like to add a variable of efficiency
# basically how much can you generate, with the least amount of co2
mingCo2pkWh = min(finalElectricityData2022$gCo2pkWh, na.rm = TRUE)
maxgCo2pkWh = max(finalElectricityData2022$gCo2pkWh, na.rm = TRUE)

finalElectricityData2022 = finalElectricityData2022 %>%
  mutate(EfficiencyScore = 100 * (1 - (gCo2pkWh - mingCo2pkWh) / (maxgCo2pkWh - mingCo2pkWh))) %>%
  select(State, EfficiencyScore, gCo2pkWh, netKWhGeneration, gCo2, millionTonsOfCO2, AverageRetailPrice
         TotalRetailSales.MWh., NetSummerCapacity.MW., NetGeneration.MWh.)

head(finalElectricityData2022)
```

```
##        State EfficiencyScore    gCo2pkWh netKWhGeneration       gCo2
## 1    Alabama        99.09246 0.008303814     144788893000 1202300000
## 2     Alaska        64.41126 0.067865449       6694128000  454300000
## 3    Arizona        98.94786 0.008552154     104698773000  895400000
## 4   Arkansas        97.78543 0.010548512      65905030000  695200000
## 5 California        93.65479 0.017642501     203383857000 3588200000
## 6   Colorado        94.13973 0.016809659      58044009000  975700000
##   millionTonsOfCO2 AverageRetailPrice.cents.kWh. TotalRetailSales.MWh.
## 1            109.3                         11.59              87027545
## 2             41.3                         20.73               6002080
## 3             81.4                         11.31              84196517
## 4             63.2                          9.91              48997663
## 5            326.2                         22.33             251869136
## 6             88.7                         11.75              56763041
##   NetSummerCapacity.MW. NetGeneration.MWh.
## 1                 28910          144788893
## 2                  2820            6694128
## 3                 28202          104698773
## 4                 14954           65905030
## 5                 85981          203383857
## 6                 18092           58044009
```

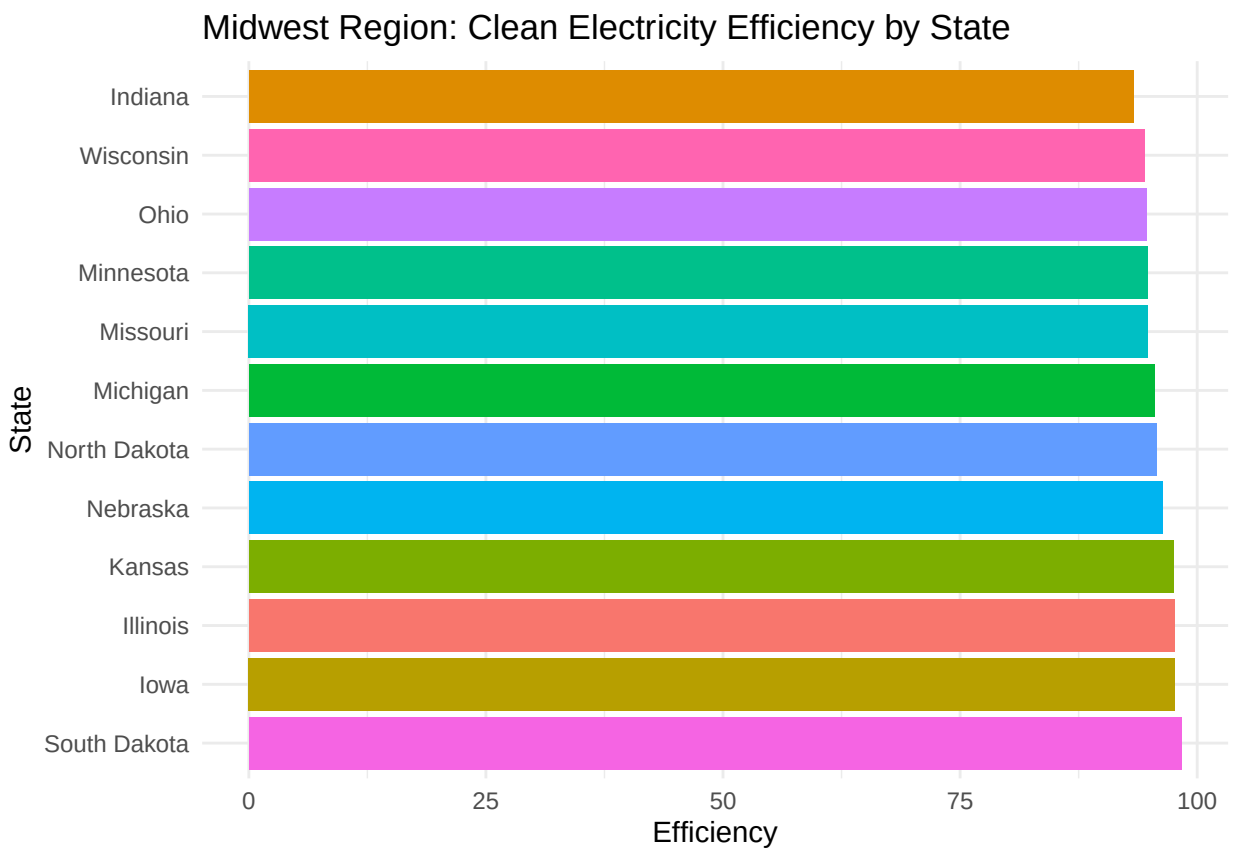
Now with this data, I can automate the efficiency based on the users input. So when the user says what state they want to see, it will fill the values for grams per mile of the electric vehicles with the formula gCo2/Kwh divided by miles/Kwh and this will show the user how electric vehicles perform in this state compared to other efficient vehicles.

```
# no we can create an efficiency scale based on the region
westEfficiencyData2022 = finalElectricityData2022 %>%
  filter(State %in% c("Arizona", "Colorado", "Idaho", "Montana", "Nevada",
                      "New Mexico", "Utah", "Wyoming", "Alaska", "California",
                      "Hawaii", "Oregon", "Washington"))

midWestEfficiencyData2022 = finalElectricityData2022 %>%
  filter(State %in% c("Illinois", "Indiana", "Michigan", "Ohio", "Wisconsin", "Iowa", "Kansas", "Minnes
                      "Nebraska", "North Dakota", "South Dakota"))
```

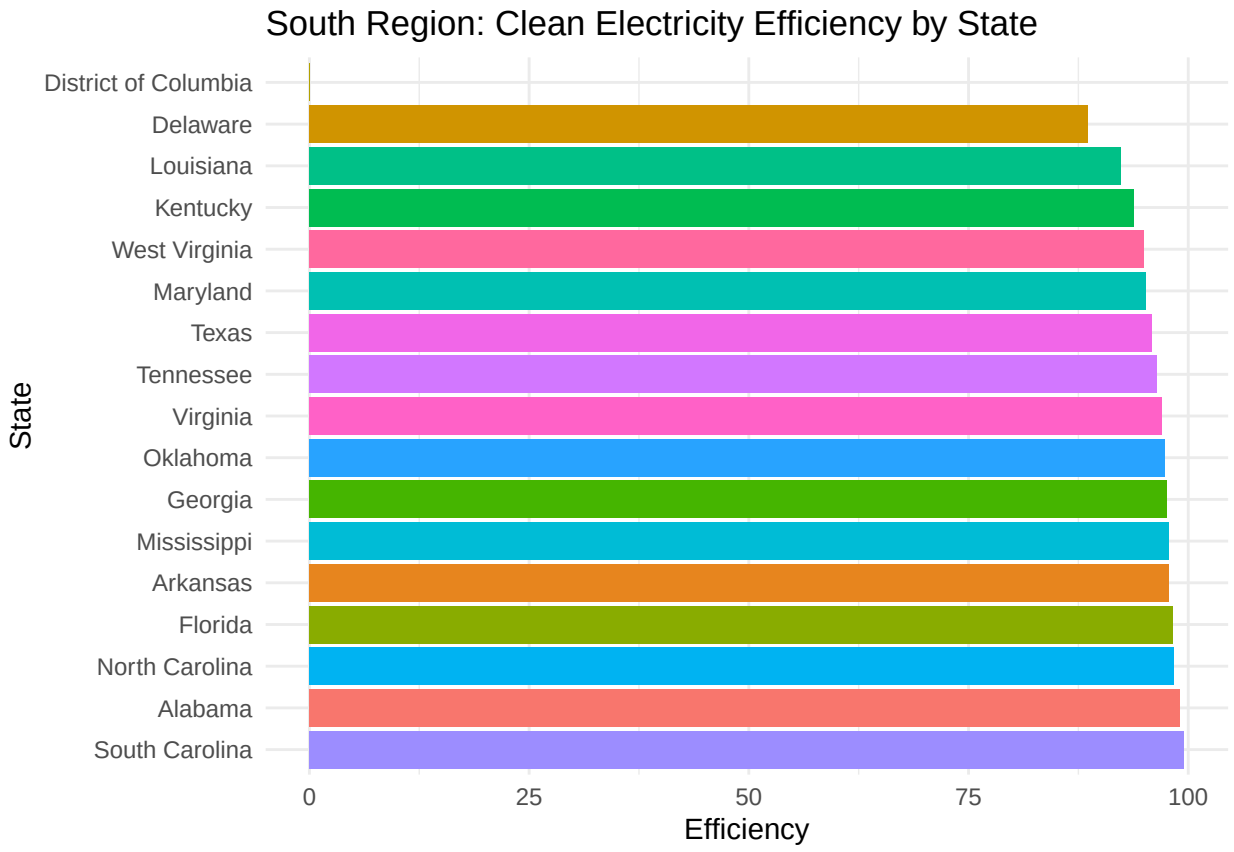
```r
southEfficiencyData2022 = finalElectricityData2022 %>%
  filter(State %in% c("Delaware", "Florida", "Georgia", "Maryland", "North Carolina", "South Carolina",
                      "West Virginia", "Alabama", "Kentucky", "Mississippi", "Tennessee", "Arkansas", "
                      "Oklahoma", "Texas", "District of Columbia"))

northEfficiencyData2022 = finalElectricityData2022 %>%
  filter(State %in% c("Connecticut", "Maine", "Massachusetts", "New Hampshire", "Rhode Island", "Vermon
                      "New Jersey", "New York", "Pennsylvania"))

efficiencyRegionPlot = function(data, region_name) {
  ggplot(data, aes(x = reorder(State, -EfficiencyScore), y = EfficiencyScore, fill = State)) +
    geom_bar(stat = "identity") +
    coord_flip() +  # Flip coordinates for better readability
    theme_minimal() +
    labs(title = paste(region_name, "Region: Clean Electricity Efficiency by State"),
         x = "State",
         y = "Efficiency") +
    theme(legend.position = "none")  # Hide legend for better visualization
}

efficiencyRegionPlot(southEfficiencyData2022, "West")
```



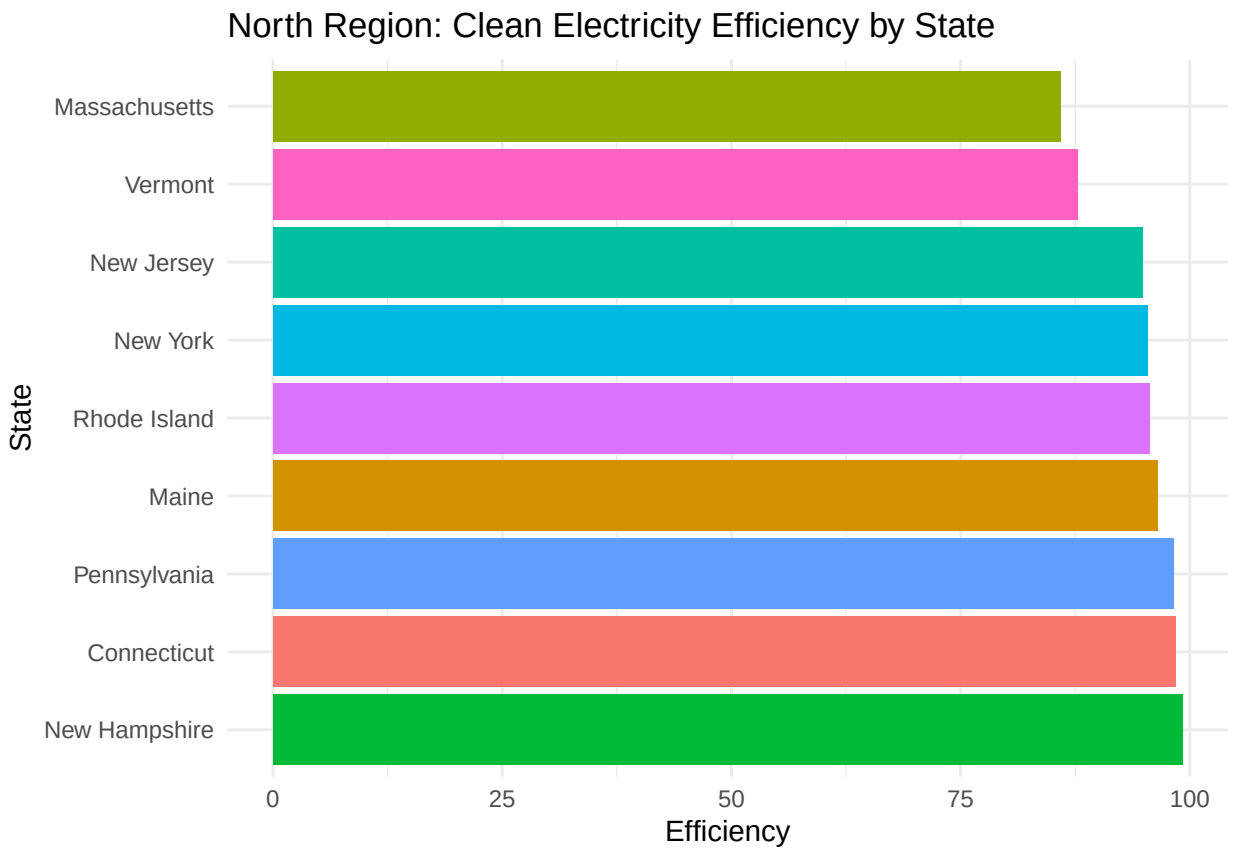
```
efficiencyRegionPlot(midWestEfficiencyData2022, "Midwest")
```




```
efficiencyRegionPlot(southEfficiencyData2022, "South")
```




```
efficiencyRegionPlot(northEfficiencyData2022, "North")
```




```
# after cleaning and manipulating the data, we will now export them as csv's to import into sql
write_csv(nonElectricVehicleData2022, "vehicleData2022.csv")
write_csv(stateData2022, "stateEmissionsData2022.csv")
write_csv(electricityGeneration2022, "stateGenerationData2022.csv")
write_csv(electricVehicles2022, "electricVehicleData2022.csv")
write_csv(finalElectricityData2022, "completeStateData2022")
```